# Shanmugam Subramanian

Mechanical Engineer II | Automation, Fixtures & Machine Design | Hyderabad, India

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## SUMMARY


Mechanical engineer with 4 years of NPD and NPI experience and in-house automation tooling for 5G antenna manufacturing. Developed a pneumatic punching fixture (2 operators to 1, yield 60% to 95%), assembly fixtures and press tools (40% less rework, 75% faster riveting), and supported contract-manufacturer setup. Built a 1:64 gearbox-driven conveyor platform and a 6-DOF robotic arm as personal projects.


## EXPERIENCE

**JMA Wireless** — Hyderabad, India
*Mechanical Engineer II* May 2023 – Present

- Developed a **pneumatic-actuated pivot punching fixture** that automated a 2-person task to one operator and **improved yield from 60% to 95%** (saving $0.4 per failed cycle).
- Automation & process-improvement by developing RF PCB sub-assembly fixtures, rivet press-pull tools, & alignment jigs, achieving **40% rework reduction, 75% faster riveting, and cutting up to 20–50 mins per antenna** in the production line.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.
- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- Released RF PCBs, sheet metal, injection-molded, die-cast, CNC, EDM, & FDM parts with a **90% FAI pass rate**.
- Built Excel-based **hand-calculation tools** to estimate drag force from geometry, crimping force for pneumatic cylinder sizing, and snap-joint behavior in plastic part design.
- Designed pole, wall, ceiling, & tri-sector mounting brackets; gravity-assisted design **halved installation manpower**.
- Authored a **utility patent draft** for an injection-molded, snap-fit PCB/sheet-metal support pillar (twist-lock base, flexing offset arms, two-cavity mold).
- Introduced topology-optimized FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.
- Executed **static structural analysis** with FOS targets of 1.5–3 (metals) & 2–5 (plastics).
- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.

**MulticoreWare India Pvt. Ltd. (Uhnder Team)** — Chennai, India
*Mechanical Design Engineer (R&D)* Jul 2022 – Apr 2023

- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.
- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.

**MulticoreWare India Pvt. Ltd. (Uhnder Team)** — Chennai, India
*Mechanical Project Intern* Mar 2022 – Jun 2022

- Created company-wide central library in SolidWorks with 700+ frequently used fasteners, reducing design lead time.
- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

## EDUCATION

**Bachelor of Engineering** — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

## PROJECTS AND COMPETITIONS

**Gearbox design** — Dec 2025 Personal project
Designed and built a 1:64 multi-stage spur gear gearbox system to power a conveyor platform for scaled vehicle testing.

**6-DOF Mimicking Robot**— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.

**Hole Anchor Gripper** Personal project
Designed a cam-lever expanding anchor gripper that inserts into holes and locks to lift loads.

**Prodigy Design Competition**— May 2021 NIT Trichy
Designed a regenerative adhesive rolling-belt mechanism for water-body waste removal within a 24-hour time frame; delivered CAD models & documentation, securing runner-up position among 50 teams.

## SKILLS & SOFTWARE

**Design:** SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing, **jigs, fixtures, pneumatic & press-tool design**, gear & drivetrain design
**Simulation:** Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis, hand-calculation tools (Excel)
**Programming & Embedded systems:** Python, MATLAB Simulink & Simscape, C, C++, Arduino
**Other Tools:** Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM, BOM costing, tooling transfer